

OFFICIAL TECHNICAL GUIDE

Matrix-IoT Hands-On Integration Tutorial

Target Audience: Engineering Students, Lab Instructors, System Engineers | matrix-iot.com



Tutorial Objective

By completing this tutorial, you will configure an ESP32 micro-controller with an RFID scanner, connect it to the Matrix-IoT Gateway via MQTT/TLS, and automatically push real-time student attendance and lab assessment scores into Matrix Cloud LMS.

MODULE 1

Hardware Assembly & Pin Wiring

Connect your ESP32 NodeMCU dev kit to the PN532 RFID NFC reader using the SPI interface according to the wiring pinout below:

Pin Mapping: ESP32 to PN532 RFID Module

ESP32 Pin	PN532 RFID Module Pin	Description / Signal
3V3	VCC	3.3V Power Supply
GND	GND	Ground Reference
GPIO 18	SCK	SPI Clock
GPIO 19	MISO	SPI Master In Slave Out
GPIO 23	MOSI	SPI Master Out Slave In
GPIO 5	SS / CS	SPI Chip Select

MODULE 2

ESP32 MicroPython Firmware

Upload the following MicroPython code snippet to your ESP32 device. This firmware initializes Wi-Fi, scans for student RFID cards, and publishes JSON payloads to Matrix-IoT Broker.

```
# matrix_iot_node.py - Matrix-IoT ESP32 MicroPython Client
import network
import ujson
import time
from umqtt.simple import MQTTClient
from machine import Pin, SPI

# 1. Network Configuration
WIFI_SSID = "College_Lab_WiFi"
WIFI_PASS = "LabSecurePass2026"
MQTT_BROKER = "mqtt.matrix-iot.com"
CLIENT_ID = "esp32_lab_station_01"
COLLEGE_TENANT_ID = "meridian_college"

def connect_wifi():
    wlan = network.WLAN(network.STA_IF)
    wlan.active(True)
    if not wlan.isconnected():
        print("Connecting to Wi-Fi...")
        wlan.connect(WIFI_SSID, WIFI_PASS)
        while not wlan.isconnected():
            time.sleep(0.5)
    print("Wi-Fi Connected! IP:", wlan.ifconfig()[0])

# 2. Main Scan & Publish Loop
def publish_attendance(rfid_uid):
    client = MQTTClient(CLIENT_ID, MQTT_BROKER, port=8883, ssl=True)
    client.connect()

    payload = ujson.dumps({
        "tenant_id": COLLEGE_TENANT_ID,
        "station_id": CLIENT_ID,
        "rfid_card_uid": rfid_uid,
        "timestamp": time.time(),
        "action": "LAB_CHECK_IN"
    })
```

```
topic = "matrix-iot/" + COLLEGE_TENANT_ID + "/attendance"
client.publish(topic, payload)
print("Published RFID UID [" + rfid_uid + "] to Matrix Cloud LMS!")
client.disconnect()

connect_wifi()
# Simulated RFID scan callback
publish_attendance("9A-4F-88-C2")
```

MODULE 3

Matrix LMS API Token & Webhook Verification

In your Matrix LMS College Admin portal, generate an IoT Device API Key under `/settings/api` . Send a HTTP POST request to verify edge authentication:

```
curl -X POST https://api.matrix-iot.com/v1/telemetry/sync \
-H "Content-Type: application/json" \
-H "Authorization: Bearer mtx_live_99812a4f781c" \
-d '{
  "tenant_id": "meridian_college",
  "course_id": "CS302-IOT",
  "student_rfid": "9A-4F-88-C2",
  "lab_bench_id": "BENCH_04",
  "hardware_status": "PASSED"
}'
```

MODULE 4

Automated Assessment & Grade Synchronization

When student hardware tests complete, Matrix-IoT triggers an automated lab grade update directly in the student's Gradebook in Matrix Cloud LMS.

 Result Verification

Navigate to **Matrix LMS > Attendance** or **Assessments** to view live updated timestamp records, lab duration metrics, and verified grade submissions.